# Robust LLM-Based Beam Prediction: Alignment-Stage Reliability for 6G V2X

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***Abstract*—Reliable sensing-aided beam prediction for mmWave/THz V2X communications requires addressing three coupled failure modes that arise in real deployments but are largely absent from existing evaluation protocols: asynchronous multi-sensor pipelines with heterogeneous latency and staleness, time-varying modality reliability under weather, occlusion, and sensor faults, and prohibitive labeling cost when beam labels rely on exhaustive sweeping. We identify a pervasive *synchronized evaluation bias*—the practice of benchmarking beam predictors only on perfectly aligned modality tuples—which conceals critical performance cliffs in practice and argue that the correct evaluation target is *deployment-realistic robustness*.**

**This paper makes the following contributions. First, we develop a delay-regime-aware asynchronous alignment mechanism that combines windowed candidate retrieval, staleness-aware temporal weighting, first-order feature-space repair, and residual refinement with regime-conditioned adapters. We provide a staleness-induced beam misalignment bound showing that the probability of selecting a suboptimal beam grows proportionally with feature velocity and delay, and that windowed staleness-aware weighting provably reduces the expected alignment error. Second, we propose reliability-guided fusion with reliability-weighted alignment regularization, grounding the reliability score in an uncertainty-proxy formulation rather than a black-box gate. We formally derive a gradient contamination bound demonstrating that standard unweighted alignment allows noise variance to directly inflate the alignment gradient, and that precision-based reliability weighting suppresses this contamination exponentially. The central conceptual contribution is that noise must be prevented from propagating *at the alignment stage*—not merely at the fusion output—a fundamental vulnerability overlooked in prior work. Third, we present a label-efficient two-stage training strategy with degradation-aware MoE-LoRA adaptation. Fourth, we provide a detailed complexity analysis confirming real-time feasibility under V2X latency constraints. Finally, we define a reproducible robustness evaluation suite including tail-risk metrics and systematic ablations that make every claim independently falsifiable.**

***Index Terms*—6G V2X, mmWave/THz, beam prediction, multimodal sensor fusion, large language models, robustness, missing modalities, asynchronous fusion, PEFT, LoRA, MoE, reliability estimation, gradient contamination.**

## I. Introduction

Millimeter-wave (mmWave) and terahertz (THz) V2X links promise multi-Gbps throughput but rely on narrow beams that must be re-aligned frequently under mobility and blockage. Exhaustive beam sweeping is therefore a major overhead and latency bottleneck, motivating learning-based beam prediction that leverages out-of-band sensing context—camera, LiDAR/radar, GPS/IMU—to infer the best beam with minimal sweeping. Large-scale real-world multimodal datasets such as DeepSense 6G [1] and real-world demonstrations of perception-aided prediction [2], [3] have significantly accelerated this direction.

Despite this progress, robust deployment remains challenging because the dominant failure modes in practice are not insufficient model capacity but *misaligned and unreliable streams*. Modern V2X stacks exhibit three characteristic failure modes: *(i) sensor asynchrony*—heterogeneous processing delays between camera and point cloud pipelines, networking-induced frame drops, and decision-time temporal mismatch across modalities; *(ii) modality reliability variation*—different sensors fail differently under weather, occlusion, and hardware faults, and naive fusion propagates corrupted modality noise into shared representations; *(iii) label scarcity*—beam labels typically require exhaustive sweeping or dense ground-truth acquisition, making repeated adaptation across deployment scenarios prohibitively expensive.

A fundamental methodological problem compounds these difficulties. We use the term **synchronized evaluation bias** to describe the widespread practice of evaluating beam predictors only on perfectly synchronized modality tuples. This convention hides critical performance cliffs that emerge in real deployments where sensor streams are asynchronous and partially missing. We argue that **deployment-realistic robustness**—maintaining target beam accuracy under the joint occurrence of asynchrony, modality degradation, and domain shift at operationally relevant levels—is the correct evaluation target for V2X beam prediction.

Recent work has begun exploring large language models (LLMs) for beam prediction by tokenizing perception and temporal signals and framing beam selection as sequence modeling [4]–[6]. LLM-MM demonstrates that parameter-efficient MoE-LoRA adaptation with lightweight preprocessing can substantially improve robustness to weather and traffic variations with minimal trainable parameters [7]. Nevertheless, two gaps remain for practical end-to-end deployment. First, most pipelines assume synchronized modality tuples and do not systematically account for staleness or heterogeneous delay regimes. Second, robustness is typically enforced *after* fusion, whereas corrupted modalities can poison learning earlier, through the alignment regularization and cross-modal coupling that occur during training.

**Why an LLM backbone?** The choice of an LLM over a standard multimodal Transformer is justified by four complementary properties. *Beam prediction as ranking and sequence modeling:* Selecting a beam from a large codebook is fundamentally a ranking problem conditioned on multimodal context. Pretrained LLMs are designed for sequential token prediction and excel at contextual reasoning over long token sequences, making them a natural fit for this task. *Long-horizon temporal reasoning:* Under asynchrony and intermittent sensor availability, effective prediction requires integrating

information over extended temporal histories—a regime in which LLMs, with their large context windows and pretrained sequence modeling, outperform shallow Transformers. *PEFT compatibility:* LLMs provide a natural interface for parameter-efficient fine-tuning. LoRA and MoE-LoRA modules [8], [9] attach to frozen pretrained weights, enabling rapid few-label adaptation without retraining from scratch—critical when beam labels are scarce and deployment environments shift. *Deployment adaptability:* The tokenization paradigm unifies heterogeneous modality inputs—images, point clouds, scalars—into a common sequence representation. Combined with PEFT, a single pretrained backbone can serve multiple deployment configurations via lightweight adapter switching.

**Conceptual contribution.** The central insight of this work is that the alignment stage—where cross-modal representations are encouraged to agree—is the most critical vulnerability point for noise propagation in multimodal fusion. Prior robust fusion methods apply gating or attention at the fusion output, but this cannot prevent corrupted modalities from contaminating the shared representation space during training via alignment losses. We elevate this observation from an engineering design choice to a *conceptual principle*: reliability must be enforced at the alignment stage, not only downstream. This principle is formalized through a gradient contamination bound (Section III-B) and a staleness-induced beam misalignment bound (Section III-A).

The paper is organized as follows. Section II presents the proposed framework. Section III provides theoretical analysis. Section IV analyzes computational complexity. Section V describes the experimental evaluation. Section VI discusses related work and the identified gap.

## II. Proposed Framework

The inference pipeline at decision time $t$ proceeds as follows: (i) per-modality observations with timestamps are collected; (ii) delay-aware aligned modality features are computed; (iii) modality reliability is estimated and features are gated; (iv) gated features are fused into a compact token sequence; (v) LLM inference is performed with PEFT/MoE adapters; (vi) a beam ranking is output. We describe each stage in turn.

### A. C1: Delay-Regime-Aware Asynchronous Alignment

Each modality $\omega \in \{\text{cam}, \text{lidar}, \text{radar}, \text{gps}\}$ arrives with a timestamp $\tau_\omega$ and staleness $\Delta t_\omega = t - \tau_\omega$. Heterogeneous processing pipelines mean that the latest frames across modalities are rarely synchronized; forcing synchronous tuples mixes inconsistent world states and degrades beam decisions. The proposed alignment mechanism addresses this through four complementary operations.

**(a) Windowed candidate retrieval.** For each modality we maintain a short buffer of the $K+1$ most recent observations $\{x_\omega(t-\tau)\}_{\tau=0}^{K}$ and encode them:

$$\mathbf{h}_\omega(t-\tau) = E_\omega(x_\omega(t-\tau)), \quad \tau \in [0, K]. \tag{1}$$

**(b) Staleness embedding and temporal weighting.** Staleness is embedded via a learnable mapping $\phi(\Delta t_\omega)$ implemented as sinusoidal time features followed by an MLP. A score for each buffer candidate is computed as

$$s_{\omega,\tau}(t) = \mathbf{v}_\omega^\top \tanh(\mathbf{W}_{h,\omega}\mathbf{h}_\omega(t-\tau) + \mathbf{W}_{d,\omega}\phi(\Delta t_\omega) + \mathbf{b}_\omega), \tag{2}$$

and the preliminary aligned feature is a normalized soft mixture:

$$\tilde{\mathbf{h}}_\omega(t) = \sum_{\tau=0}^{K} \alpha_{\omega,\tau}(t)\, \mathbf{h}_\omega(t-\tau), \quad \alpha_{\omega,\tau}(t) = \mathrm{softmax}_\tau(s_{\omega,\tau}(t)). \tag{3}$$

This explicitly learns *which past frame is most compatible with the decision time*, rather than assuming perfect synchronization.

**(c) Short-gap repair.** When a modality is absent from the buffer or too stale ($\Delta t_\omega > \Delta t_{\max}$), a lightweight continuity-based repair is applied before refinement:

$$\hat{\mathbf{h}}_\omega(t) = \tilde{\mathbf{h}}_\omega(t-1) + \eta_\omega\Big(\tilde{\mathbf{h}}_\omega(t-1) - \tilde{\mathbf{h}}_\omega(t-2)\Big), \tag{4}$$

where $\eta_\omega = \sigma(\eta_\omega^{\text{raw}}) \in [0,1]$ is a *per-modality learnable scalar* initialized to $\approx 0.5$. When $\eta_\omega = 0$, the repair reduces to a zero-order hold, which assumes feature stationarity. When $\eta_\omega = 1$, it performs first-order linear extrapolation in feature space, assuming constant velocity. Learning $\eta_\omega$ allows the model to adapt: fast-changing modalities such as camera benefit from larger $\eta_\omega$, while slowly-varying ones such as GPS prefer values close to zero.

The extrapolated feature satisfies a bounded norm guarantee. Because encoder outputs are layer-normalized and $\eta_\omega \in [0,1]$:

$$\|\hat{\mathbf{h}}_\omega(t)\| \le (1 + 2\eta_\omega)\, C_{\text{norm}} \le 3\, C_{\text{norm}}, \tag{5}$$

where $C_{\text{norm}}$ is the layer-norm bound. This prevents unbounded feature drift even under consecutive missing frames. In practice the repair is invoked only for short gaps ($\le 3$ consecutive missing frames); for longer outages, the reliability estimator in Section II-B gates the modality to near-zero.

**(d) Residual alignment refinement with regime conditioning.** A small residual network corrects the aligned feature with explicit awareness of both delay and missingness:

$$\bar{\mathbf{h}}_\omega(t) = \tilde{\mathbf{h}}_\omega(t) + \mathrm{MLP}_\omega\Big([\tilde{\mathbf{h}}_\omega(t);\, \phi(\Delta t_\omega);\, \mathbf{m}_\omega(t)]\Big), \tag{6}$$

where $\mathbf{m}_\omega(t)$ is a binary indicator of whether the modality was repaired or missing. To handle heterogeneous deployment latency profiles, staleness is discretized into regimes $r_\omega(t) = \mathrm{bin}(\Delta t_\omega)$ (e.g., low $[0, 50)$ ms, medium $[50, 150)$ ms, high $[150, \infty)$ ms), and regime-specific lightweight adapters are activated inside $\mathrm{MLP}_\omega$:

$$\mathrm{MLP}_\omega(\cdot) \Leftarrow \mathrm{MLP}_\omega(\cdot;\, \theta_\omega^{(r_\omega)}), \tag{7}$$

enabling alignment behavior to specialize to different latency budgets without full retraining.

### B. C2: Reliability-Guided Fusion with Reliability-Weighted Alignment Regularization

Under weather, occlusion, or sensor faults, naive fusion allows noise from degraded modalities to corrupt shared representations. The approach in this section prevents noise propagation at two coupled points: the alignment regularization during training and the fusion gating during inference.

**(a) Reliability estimation: uncertainty-proxy formulation.** Rather than treating reliability as a black-box learned gate $w_\omega = \sigma(g_\omega(\cdot))$, we ground it in a principled uncertainty proxy. The encoder is augmented with a variance head that produces a scalar log-variance alongside the feature mean:

$$\log \hat{\sigma}^2_\omega(t) = f^{\text{var}}_\omega\big([\bar{\mathbf{h}}_\omega(t);\, \mathbf{q}_\omega(t);\, \mathbf{m}_\omega(t)]\big)\,, \tag{8}$$

where $\mathbf{q}_\omega(t)$ are available quality cues such as missing-rate indicator, LiDAR point density, and GPS jitter proxy. The reliability score is the precision-based weight:

$$w_\omega(t) = \frac{\exp(-\hat{\sigma}^2_\omega(t))}{\sum_{\omega'} \exp(-\hat{\sigma}^2_{\omega'}(t))}. \tag{9}$$

This assigns high weight to modalities with low estimated feature uncertainty and low weight to noisy or degraded ones. The variance head is trained jointly with the main objective so that the $\hat{\sigma}^2_\omega$ terms appearing in the alignment loss and beam loss provide a self-supervised calibration signal.

To verify that $\hat{\sigma}^2_\omega$ constitutes a meaningful quantity rather than a learned artifact, we propose three calibration diagnostics. First, the Pearson correlation $\rho(\hat{\sigma}^2_\omega, e_\omega)$ between estimated variance and the per-modality beam prediction error $e_\omega$ (measured by removing all other modalities) should be strongly positive ($\rho > 0.5$). Second, reliability-conditioned accuracy—binning samples by $w_\omega(t)$ and reporting Top-1 accuracy per bin—should show significantly higher accuracy in high-$w_\omega$ bins. Third, under controlled degradation at severity $\alpha$, $\hat{\sigma}^2_\omega$ should increase monotonically with $\alpha$ for each modality.

**(b) Coarse modality gating.** Aligned features are gated by reliability before fusion:

$$\mathbf{h}^{\text{gated}}_\omega(t) = w_\omega(t) \cdot \bar{\mathbf{h}}_\omega(t). \tag{10}$$

**(c) Reliability-weighted alignment regularization.** The key novelty is that reliability weights are applied directly inside the alignment loss, suppressing contaminated gradients at their source during training:

$$\mathcal{L}_{\text{align}} = \sum_t \sum_{\omega \neq \omega'} \underbrace{w_\omega(t)\, w_{\omega'}(t)}_{\text{suppress noisy pairs}} \left\| \bar{\mathbf{h}}_\omega(t) - P_{\omega \leftarrow \omega'}\big(\bar{\mathbf{h}}_{\omega'}(t)\big) \right\|^2_2, \tag{11}$$

where $P_{\omega \leftarrow \omega'}(\cdot)$ is a lightweight projection between modality feature spaces. When either modality is unreliable the alignment term is suppressed, preventing degraded representations from dragging the shared feature space. The theoretical justification for this design is provided in Section III-B.

**(d) Fusion and LLM interface.** Gated features are fused by a compact cross-attention module $\mathcal{F}(\cdot)$ to generate $M$ tokens that serve as a soft prompt/prefix conditioning the LLM:

$$\mathbf{Z}(t) = \mathcal{F}\big(\{\mathbf{h}^{\text{gated}}_\omega(t)\}_\omega\big) \in \mathbb{R}^{M \times d}. \tag{12}$$

### C. C3: Label-Efficient Two-Stage Training with PEFT

Beam labels require exhaustive sweeping, and label distributions shift across environments, making full fine-tuning of large backbones costly and brittle for repeated adaptation. We address this through a hierarchical two-stage training strategy.

**Stage 1: Self-supervised multimodal pretraining.** Encoders and the fusion module are pretrained on abundant unlabeled sequences to learn stable geometry and motion representations that are robust to missingness and asynchrony. Three complementary self-supervised objectives are used:

$$\mathcal{L}_{\text{MR}} = \sum_{t,\omega} \left\| D_\omega(\text{Mask}(\mathbf{h}_\omega(t))) - \mathbf{h}_\omega(t) \right\|^2_2, \tag{13}$$

$$\mathcal{L}_{\text{CM}} = \sum_{t,\omega} \left\| Q_\omega(\{\mathbf{h}_{\omega'}(t)\}_{\omega' \neq \omega}) - \mathbf{h}_\omega(t) \right\|^2_2, \tag{14}$$

$$\mathcal{L}_{\text{TF}} = \sum_{t} \left\| F(\mathbf{Z}(t)) - \mathbf{Z}(t+1) \right\|^2_2, \tag{15}$$

denoting masked reconstruction (T1), cross-modal prediction (T2), and short-horizon temporal forecasting (T3), respectively. The total Stage 1 objective is $\mathcal{L}_{\text{SSL}} = \lambda_{\text{MR}}\mathcal{L}_{\text{MR}} + \lambda_{\text{CM}}\mathcal{L}_{\text{CM}} + \lambda_{\text{TF}}\mathcal{L}_{\text{TF}}$.

The alignment loss $\mathcal{L}_{\text{align}}$ in (11) is *not* active in Stage 1. This is because the reliability weights $w_\omega$ are co-trained with the beam prediction objective; without beam-label supervision they lack a meaningful gradient signal and risk degenerating to trivial values. The cross-modal prediction task (T2) provides an implicit alignment signal by requiring each modality to predict the others, serving as a warm initialization for the explicit reliability-weighted alignment in Stage 2.

**Stage 2: Supervised fine-tuning with PEFT.** A beam prediction head is attached to the pooled LLM output $\mathbf{o}(t)$. For a codebook of size $B$:

$$\hat{\mathbf{y}}(t) = \text{softmax}(\mathbf{W}\mathbf{o}(t) + \mathbf{b}) \in \mathbb{R}^B, \quad \mathcal{L}_{\text{beam}} = \sum_t \text{CE}(\hat{\mathbf{y}}(t), \mathbf{y}(t))\,. \tag{16}$$

The LLM backbone is frozen and only LoRA modules [8], [9] and the beam head are trained.

**Degradation-aware MoE-LoRA routing.** Inspired by conditional computation [10] and the MoE-LoRA approach of LLM-MM [7], each sample is routed to one of $E$ expert adapters based on both delay regime and reliability pattern:

$$e(t) = \arg\max_{k \in \{1,\ldots,E\}} \mathcal{R}_k([\mathbf{o}(t);\, \mathbf{r}(t);\, \mathbf{w}(t)])\,, \tag{17}$$

where $\mathbf{r}(t) = [r_\omega(t)]_\omega$ and $\mathbf{w}(t) = [w_\omega(t)]_\omega$. Only the selected expert's LoRA parameters are activated at inference, enabling experts to specialize to distinct failure-mode patterns (e.g., camera unreliable + LiDAR reliable; GPS drift; high-delay regime).

The Stage 2 training objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{beam}} + \lambda_{\text{align}}\mathcal{L}_{\text{align}}, \tag{18}$$

where $\mathcal{L}_{\text{align}}$ is reliability-weighted per (11), ensuring degraded modalities do not poison alignment during supervised fine-tuning.

## III. Theoretical Analysis

We formalize the two core design principles—delay-aware weighting and reliability-weighted alignment—as theoretically grounded claims. Section III-A bounds the beam misalignment probability as a function of staleness, justifying the alignment mechanism in Section II-A. Section III-B bounds the gradient contamination induced by noisy modalities under standard alignment, justifying the reliability-weighted formulation in Section II-B.

### A. Staleness-Induced Beam Misalignment Bound

**Setup.** Consider modality $\omega$ observed at time $\tau_\omega$ with staleness $\Delta t_\omega = t - \tau_\omega$. Let $\mathbf{s}_\omega(t) \in \mathbb{R}^d$ denote the true latent state of modality $\omega$ at decision time $t$, and assume bounded feature velocity:

$$\|\mathbf{s}_\omega(t) - \mathbf{s}_\omega(\tau_\omega)\| \le v_\omega \cdot \Delta t_\omega, \tag{19}$$

where $v_\omega$ is a modality-dependent Lipschitz constant ($v_{\text{cam}}$ is large under high mobility; $v_{\text{gps}}$ is small). For the purpose of theoretical analysis, we model the fused state as a weighted combination of per-modality states:

$$\mathbf{s}(t) = \sum_\omega \gamma_\omega \, \mathbf{s}_\omega(t), \tag{20}$$

where $\gamma_\omega \ge 0$ are fixed fusion weights satisfying $\sum_\omega \gamma_\omega = 1$, used here as bounded constants for the purpose of deriving the upper bound. These play the role of a linearized readout of the cross-attention fusion module $\mathcal{F}(\cdot)$ defined in Section II-B and are distinct from the online reliability weights $w_\omega(t)$.

**Misalignment bound.** Let $b^*(t) = \arg\max_b \mathbf{a}_b^\top \mathbf{s}(t)$ be the optimal beam at time $t$, where $\mathbf{a}_b$ is the beam steering vector. Using stale per-modality observations $\mathbf{s}_\omega(\tau_\omega)$ instead of $\mathbf{s}_\omega(t)$ introduces a beam score error bounded by

$$|\mathbf{a}_b^\top (\mathbf{s}(t) - \mathbf{s}(\tau))| \le \|\mathbf{a}_b\| \cdot \sum_\omega \gamma_\omega v_\omega \Delta t_\omega, \tag{21}$$

where $\mathbf{s}(\tau) = \sum_\omega \gamma_\omega \mathbf{s}_\omega(\tau_\omega)$ denotes the stale fused state. Consequently, the probability of selecting a suboptimal beam satisfies

$$P_\xi\Big(\hat{b} \ne b^*(t)\Big) \le P_\xi\left(\max_{b \ne b^*} |\mathbf{a}_{b^*}^\top \mathbf{s}(t) - \mathbf{a}_b^\top \mathbf{s}(t)| \le 2 \sum_\omega \gamma_\omega v_\omega \Delta t_\omega\right), \tag{22}$$

where $P_\xi(\cdot)$ denotes probability over the test-time scene distribution $\xi$, which induces randomness in the true state $\mathbf{s}(t)$ and hence in the beam score margin $\max_{b \ne b^*} |\mathbf{a}_{b^*}^\top \mathbf{s}(t) - \mathbf{a}_b^\top \mathbf{s}(t)|$. The bound states that misalignment probability is governed by whether the staleness-induced score perturbation $2\sum_\omega \gamma_\omega v_\omega \Delta t_\omega$ exceeds the scene-dependent beam score margin; in deployment scenarios with small margin (e.g., closely spaced beams under high mobility), even moderate staleness can cause misalignment.

**Effect of windowed staleness-aware weighting.** Under the weighting mechanism of Section II-A(b), the expected alignment error after soft aggregation satisfies

$$\mathbb{E}\Big[\|\tilde{\mathbf{h}}_\omega(t) - \mathbf{s}_\omega(t)\|^2\Big] \le \min_{\tau \in [0,K]} \|\mathbf{h}_\omega(t-\tau) - \mathbf{s}_\omega(t)\|^2 + \mathcal{O}\big(\tfrac{1}{K}\big), \tag{23}$$

where the first term is the best-candidate error over the buffer and the second term is a softmax approximation overhead. To see why the overhead is $\mathcal{O}(1/K)$: the softmax with scores $s_{\omega,\tau}$ approximates a hard $\arg\min$ over $K+1$ candidates. The gap between the soft mixture and the best candidate satisfies

$$\begin{aligned}\|\tilde{\mathbf{h}}_\omega(t) - \mathbf{h}_\omega(t-\tau^*)\| &= \left\| \sum_\tau \alpha_{\omega,\tau} \left(\mathbf{h}_\omega(t-\tau) - \mathbf{h}_\omega(t-\tau^*)\right) \right\| \\ &\le \sum_{\tau \ne \tau^*} \alpha_{\omega,\tau} \cdot \|\mathbf{h}_\omega(t-\tau) - \mathbf{h}_\omega(t-\tau^*)\|,\end{aligned} \tag{24}$$

where $\tau^* = \arg\min_\tau \|\mathbf{h}_\omega(t-\tau) - \mathbf{s}_\omega(t)\|$. As the score $s_{\omega,\tau^*}$ is trained to dominate the other scores, the off-best weights $\sum_{\tau \ne \tau^*} \alpha_{\omega,\tau}$ concentrate around a residual mass that scales as $\mathcal{O}(1/K)$ in the uniform-score baseline (each of the $K$ non-optimal candidates contributes weight $\sim 1/K$), and the encoder outputs are bounded by $C_{\text{norm}}$ under layer normalization, so the gap in (24) is $\mathcal{O}(C_{\text{norm}}/K)$. In practice the learned scores sharpen this further, so $\mathcal{O}(1/K)$ is a conservative bound. Compared to simply using the latest frame, which incurs error $v_\omega^2 \Delta t_\omega^2$ from (19), the weighted combination exploits whichever buffer frame is temporally closest to the true state, reducing expected alignment error.

**Remark 1.** *Bounds* (22) *and* (23) *together establish that (i) staleness directly increases beam misalignment risk in proportion to $v_\omega \Delta t_\omega$, and (ii) windowed staleness-aware weighting provably reduces this risk compared to using only the latest observation, motivating the mechanism in Section II-A(b).*

### B. Gradient Contamination Bound

We provide a formal justification for reliability-weighted alignment by analyzing how unreliable modalities contaminate learning dynamics when standard unweighted alignment is used.

**Problem formulation.** Let the aligned feature of modality $\omega$ decompose as

$$\bar{\mathbf{h}}_\omega = \mathbf{s}_\omega + \boldsymbol{\epsilon}_\omega, \tag{25}$$

where $\mathbf{s}_\omega \in \mathbb{R}^d$ is the true latent representation and $\boldsymbol{\epsilon}_\omega \sim \mathcal{N}(\mathbf{0}, \sigma_\omega^2 \mathbf{I}_d)$ captures total degradation: $\sigma_\omega^2 \approx 0$ for clean modalities and $\sigma_\omega^2 \gg 1$ for severely corrupted ones. Consider the standard unweighted pairwise alignment loss for a single pair $(\omega, \omega')$, with the projection $P_{\omega \leftarrow \omega'}$ set to identity for clarity:

$$\mathcal{L}_{\text{align}}^{\text{std}} = \|\bar{\mathbf{h}}_\omega - \bar{\mathbf{h}}_{\omega'}\|_2^2. \tag{26}$$

**Noise contamination in the expected loss.** Taking the expectation:

$$\mathbb{E}[\mathcal{L}_{\text{align}}^{\text{std}}] = \|\mathbf{s}_\omega - \mathbf{s}_{\omega'}\|_2^2 + d\,(\sigma_\omega^2 + \sigma_{\omega'}^2). \tag{27}$$

The noise term $d(\sigma_\omega^2 + \sigma_{\omega'}^2)$ inflates the alignment objective independently of the true representation distance. This has two harmful consequences: the gradient is biased by noise, and the gradient variance increases with noise, destabilizing optimization. Concretely, the gradient through encoder parameters $\theta_\omega$ satisfies

$$\mathrm{Var}\big[\|\nabla_{\theta_\omega} \mathcal{L}_{\text{align}}^{\text{std}}\|\big] = \Theta\big((\sigma_\omega^2 + \sigma_{\omega'}^2) \cdot \|\nabla_{\theta_\omega} \bar{\mathbf{h}}_\omega\|^2\big). \tag{28}$$

We term this effect **gradient contamination**: when either modality is noisy, the alignment gradient pushes representations in unreliable directions, corrupting the shared feature space.

**Suppression under reliability-weighted alignment.** Under the precision-based weighting $w_\omega = \exp(-\hat{\sigma}_\omega^2)/Z$ defined in (9), the weighted loss is $\mathcal{L}_{\text{align}}^{\text{rw}} = w_\omega w_{\omega'} \|\bar{\mathbf{h}}_\omega - \bar{\mathbf{h}}_{\omega'}\|_2^2$. Assuming well-calibrated estimates $\hat{\sigma}_\omega^2 = \sigma_\omega^2$, the expected weighted loss becomes

$$\mathbb{E}[\mathcal{L}_{\text{align}}^{\text{rw}}] = \frac{\exp(-\sigma_\omega^2 - \sigma_{\omega'}^2)}{Z^2} \left( \|\mathbf{s}_\omega - \mathbf{s}_{\omega'}\|_2^2 + d(\sigma_\omega^2 + \sigma_{\omega'}^2) \right). \tag{29}$$

The factor $\exp(-\sigma_\omega^2 - \sigma_{\omega'}^2)$ exponentially suppresses the noise contribution as degradation increases.

**Proposition 1** (Gradient Contamination Bound)**.** *Assume the following two conditions hold:* (i) Deterministic Jacobian: *the Jacobian $\nabla_{\theta_\omega} \bar{\mathbf{h}}_\omega$ is treated as approximately deterministic with respect to the noise realizations $\boldsymbol{\epsilon}_\omega$, i.e., variance is computed only over the stochastic noise terms.* (ii) Fixed reliability weights: *the reliability weights $w_\omega$ are treated as fixed constants given calibrated estimates $\hat{\sigma}_\omega^2 = \sigma_\omega^2$, so they do not contribute additional variance in the gradient.*

*Under these conditions, let $G^{\text{std}}$ and $G^{\text{rw}}$ denote the gradient vectors of $\mathcal{L}_{\text{align}}^{\text{std}}$ and $\mathcal{L}_{\text{align}}^{\text{rw}}$ with respect to $\theta_\omega$, respectively, under precision-based reliability weighting $w_\omega = \exp(-\hat{\sigma}_\omega^2)/Z$. Then:*

$$\frac{\text{Var}[\|G^{\text{rw}}\|]}{\text{Var}[\|G^{\text{std}}\|]} \leq \frac{\exp\left(-2(\sigma_\omega^2 + \sigma_{\omega'}^2)\right)}{Z^2}. \tag{30}$$

**Remark 2** (On assumption (i))**.** *The deterministic-Jacobian assumption is standard in first-order stochastic gradient analysis when the parameter update is dominated by the loss gradient rather than higher-order noise-parameter interactions. It holds exactly when the encoder is linear in $\theta_\omega$, and is a common approximation for deep networks under mild regularity conditions.*

**Remark 3** (On assumption (ii))**.** *Treating $w_\omega$ as fixed is justified when the reliability estimator is well-calibrated and converges on a time scale slower than individual gradient steps, which is ensured by the joint but staged training described in Section II-C.*

**Remark 4** (Applicability boundary)**.** *When the reliability head and the encoder are strongly coupled in early training and $w_\omega$ fluctuates rapidly, the bound in* (30) *is best interpreted as characterizing the* asymptotic trend *of gradient contamination suppression rather than a tight per-step guarantee. Accordingly, we complement the bound with empirical validation in Experiments E1 (Section V-C) and E4 (Section V-F), which directly measure gradient norm distributions and reliability calibration under controlled degradation.*

When both modalities are clean ($\sigma_\omega^2 \approx 0$), the bound approaches $1/Z^2 \leq 1$, preserving the full alignment signal. When either modality is severely degraded ($\sigma_\omega^2 \gg 1$), the bound decays *exponentially*, effectively zeroing out the contaminated gradient.

TABLE I
COMPLEXITY OF PROPOSED MODULES. LLM BACKBONE: 7B PARAMETERS (FROZEN).

| Module | Params | FLOPs/sample | Memory |
|---|---|---|---|
| Staleness emb. + weighting | $4\times0.07$M | $4\times0.14$M | 1.1 MB |
| Short-gap repair | $4\times1$ scalar | negl. | negl. |
| Residual align. MLP | $4\times0.13$M | $4\times0.26$M | 2.1 MB |
| Regime adapters ($R$=3) | $4\times3\times0.03$M | $4\times0.06$M | 1.4 MB |
| Reliability var. head | $4\times0.03$M | $4\times0.06$M | 0.5 MB |
| Cross-attn. fusion | 0.26M | 0.52M | 1.0 MB |
| MoE-LoRA ($E$=4, $r$=16) | $4\times2dr$=0.13M | $2dr$=0.03M | 0.5 MB |
| Beam head | $Bd$=0.02M | 0.02M | 0.1 MB |
| **Total trainable** | **≈1.5M** | **≈2.7M** | **≈6.7 MB** |
| **Fraction of backbone** | **0.02%** | **<0.01%** | **<0.05%** |

**Remark 5** (Superiority over output-only gating)**.** *Post-fusion gating suppresses noisy modality contributions in the forward pass, but the alignment loss* (11) *is computed before fusion and generates gradients that flow back through both modality encoders. Output gating cannot prevent these contaminated gradients. Reliability-weighted alignment suppresses contamination at its source, providing a strictly stronger robustness guarantee during training.*

## IV. COMPLEXITY ANALYSIS

V2X beam management operates under latency budgets typically below 10 ms. Table I provides a per-module breakdown of trainable parameters, FLOPs per sample, and memory overhead relative to the frozen 7B LLM backbone, using reference dimensions $|\Omega| = 4$, $d = 256$, $K = 3$, $R = 3$, $E = 4$, $r = 16$, $B = 64$.

The total trainable parameter count is approximately 1.5M, which is 0.02% of the frozen 7B backbone. The memory footprint of all proposed modules combined is approximately 6.7 MB. Due to sparse MoE activation, only one expert is active per sample at inference, so the MoE-LoRA overhead is equivalent to a single LoRA adapter ($2dr = 8192$ parameters per layer) regardless of the total number of experts $E$. The complete set of adapters totals fewer than 0.3M parameters, enabling on-device storage and adapter switching between deployment configurations without reloading the backbone.

## V. EXPERIMENTAL EVALUATION

### A. Dataset and Evaluation Protocol

Experiments are conducted on the DeepSense 6G dataset [1], which provides synchronized multi-modal sensing data (camera, LiDAR, radar, GPS) with beam-index labels across diverse outdoor V2X scenarios. Simulation-based ablations leverage DeepMIMO [11] and ViWi [12] for controlled reproducibility.

All experiments are repeated with $\geq 5$ random seeds. Results are reported with 95% confidence intervals, and pairwise comparisons use the Wilcoxon signed-rank test. Primary beam-index metrics are Top-1 and Top-3 accuracy. Communication-relevant metrics (normalized received power, overhead proxy) are reported following [2], [3].

To overcome synchronized evaluation bias, evaluation is conducted under three controlled degradation axes:

**S1 – Asynchrony stress.** Modality-specific delays are injected at $\delta_\omega \in \{0, 50, 100, 200\}$ ms, and bursty missing intervals with burst lengths $\ell \in \{1, 3, 5, 10\}$ frames are applied to the evaluation stream. Performance is reported as degradation curves over delay magnitude and burst length.

**S2 – Modality corruption stress.** Corruption severity $\alpha \in [0, 1]$ is applied as follows: camera Gaussian blur with kernel $\sigma_{\text{blur}} = 20\alpha$ pixels; LiDAR random point dropout with probability $\alpha$; radar additive range noise $\sigma_{\text{range}} = 5\alpha$ m; GPS additive position noise $\sigma_{\text{pos}} = 50\alpha$ m. Single-modality and combined corruption ($\alpha$ applied to all modalities simultaneously) are both evaluated.

**S3 – Domain shift stress.** The model is trained on a subset of scenarios and evaluated on held-out scenarios with unseen weather and traffic conditions.

**Tail robustness metrics.** Average-case metrics are insufficient for safety-critical V2X. In addition to mean Top-1 accuracy, the following tail metrics are reported for each stress condition:

- *Worst-case bin accuracy*: test samples are binned by scenario/delay/corruption level; the accuracy of the worst-performing bin is reported.
- *Worst-10% accuracy*: samples are ranked by per-sample error; mean accuracy over the bottom 10% is reported.
- *CVaR*$_{10\%}$ [13]: $\text{CVaR}_{10\%} = \mathbb{E}[\ell \mid \ell \geq q_{90\%}]$, where $\ell$ is per-sample loss and $q_{90\%}$ is its 90th percentile.
- *Percentile degradation curves*: Top-1 accuracy at the 10th, 25th, 50th, 75th, and 90th percentiles under increasing stress.

### B. Baselines

The proposed framework is compared against the following baselines: (i) vanilla multimodal Transformer of comparable depth (24 layers, ∼350M parameters) trained from scratch with an ImageNet-pretrained vision encoder; (ii) LLM-MM [7] with the original preprocessing pipeline; (iii) an LLM backbone with naive synchronous fusion (no alignment or reliability modules). For the LLM vs. Transformer comparison (Section V-E), all methods share identical preprocessing, alignment, and reliability modules; only the backbone differs.

### C. E1: Gradient Contamination Visualization

**Purpose.** This experiment empirically validates Proposition 1 and confirms that reliability-weighted alignment suppresses gradient contamination at the source.

**Design.** Two model variants are trained under identical conditions: (i) the full model with reliability-weighted alignment, and (ii) ablation A3 (unweighted alignment, $w_\omega w_{\omega'} \to 1$). At controlled intervals during training, camera degradation is injected at severity $\alpha = 0.8$. At each injection the following quantities are recorded: the alignment gradient norm $\|\nabla_{\theta_\omega} \mathcal{L}_{\text{align}}\|$ for the degraded modality; the alignment gradient norm for each clean modality; and the representation drift $\|\bar{\mathbf{h}}_\omega^{(n)} - \bar{\mathbf{h}}_\omega^{(n-1)}\|$ between consecutive training steps $n$. Results are visualized as gradient norm distributions (violin plots) conditioned on modality reliability, with the theoretical bound curve from (30) overlaid.

**Results.**

### D. E2: Delay-Regime Specialization Effect

**Purpose.** This experiment isolates the contribution of regime-conditioned adapter specialization (Section II-A(d)) over a regime-agnostic baseline.

**Design.** Two variants are compared: regime-aware (full model with $R = 3$ regime-specific adapters: low $[0, 50)$ ms, medium $[50, 150)$ ms, high $[150, \infty)$ ms) and regime-agnostic (ablation A1, single shared alignment MLP). Both are evaluated under S1 asynchrony stress at $\delta_\omega \in \{0, 50, 100, 200\}$ ms. Results are plotted as (i) delay vs. Top-1 accuracy and (ii) delay vs. worst-10% accuracy.

**Results.**

### E. E3: LLM Necessity Verification

**Purpose.** This experiment tests whether the LLM backbone provides measurable advantages over a vanilla Transformer specifically in the regimes of label scarcity and domain shift.

**Design.** LLM + PEFT is compared against the vanilla Transformer baseline (Section V-B). Two conditions are evaluated: (i) few-label adaptation, training on $\{1\%, 5\%, 10\%, 50\%, 100\%\}$ of available beam labels and reporting Top-1 accuracy as a function of label fraction; (ii) domain shift, training on scenario A and evaluating on held-out scenario B with different weather and traffic, reporting accuracy drop.

**Results.**

### F. E4: Reliability Calibration Diagnostics

**Purpose.** This experiment verifies that the estimated variance $\hat{\sigma}_\omega^2$ constitutes a meaningful uncertainty proxy rather than a learned artifact, as required for the theoretical guarantees in Section III-B.

**Design.** For each modality, the Pearson correlation $\rho(\hat{\sigma}_\omega^2, e_\omega)$ between estimated variance and the per-modality beam prediction error $e_\omega$ is computed. Reliability-conditioned accuracy is plotted by binning test samples by $w_\omega(t)$ and reporting Top-1 accuracy per bin. Finally, under S2 corruption stress, $\hat{\sigma}_\omega^2$ is recorded as a function of $\alpha$ for each modality to verify monotonic growth.

**Results.**

### G. Systematic Ablation Study

To disentangle the contribution of each proposed component, five ablation conditions are evaluated under all three stress axes S1–S3.

- **A1: Remove delay-regime conditioning.** A single shared alignment MLP is used for all delay levels; regime-specific adapters are removed. Tests whether regime specialization provides measurable benefit beyond a single unified alignment module.

- **A2: Remove short-gap repair.** Missing frames are zero-filled: $\hat{\mathbf{h}}_\omega(t) = \mathbf{0}$. Tests whether first-order feature-space extrapolation is beneficial compared to a trivial imputation strategy.
- **A3: Remove reliability weighting from alignment.** The alignment loss uses $w_\omega w_{\omega'} \to 1$ while reliability gating at the fusion output is retained. This isolates the core claim of Proposition 1: alignment-stage protection provides robustness beyond output gating.
- **A4: Remove MoE routing.** A single shared LoRA adapter is used for all samples; failure-mode-specific expert routing is disabled. Tests whether degradation-aware routing improves adaptation over a uniform PEFT configuration.
- **A5: Replace LLM with vanilla Transformer.** The 7B LLM backbone is substituted with a 24-layer multimodal Transformer ($\sim$350M parameters). All other modules (C1–C2, PEFT head) are unchanged. Tests the LLM necessity claim under few-label and domain-shift conditions.

**Results.**

## VI. Related Work and the Identified Gap

**Sensing-aided beam prediction.** Large-scale real-world multi-modal datasets such as DeepSense 6G [1] have enabled systematic evaluation of learning-based beam prediction. Real-world demonstrations confirm that perception and semantic context improve beam tracking and prediction [2], [3], and simulation frameworks such as DeepMIMO [11] and ViWi [12] support controlled ablations. However, most published evaluation protocols implicitly assume synchronized modality tuples—what we term synchronized evaluation bias—and do not systematically address delay-induced staleness or heterogeneous pipeline latencies.

**LLM-based beam prediction.** Recent work frames beam selection as sequence modeling and tokenizes multi-modal observations as LLM inputs [4]–[6]. PEFT methods such as LoRA [8], [9] enable repeated adaptation without expensive full fine-tuning, while MoE-style sparse activation [10] supports scalable conditional computation. LLM-MM [7] demonstrates that MoE-LoRA with lightweight preprocessing can improve robustness to weather and traffic variations with minimal trainable parameters.

**Robust multimodal fusion.** Multimodal Transformers are known to be brittle under missing modalities unless robustness is explicitly designed [14]. Asynchronous streaming fusion has been studied in robotics [15], but explicit treatment of staleness and heterogeneous delay regimes in beam prediction pipelines is absent from prior work.

**The identified gap.** Even when robustness is addressed in beam prediction, it is enforced at the fusion output, leaving a fundamental vulnerability: degraded modalities can contaminate learning through cross-modal alignment losses during training. We term this the *noise propagation through alignment* failure mode. To our knowledge, no prior beam prediction work has identified this failure mode formally, provided a theoretical analysis of the induced gradient contamination, or proposed reliability-weighted alignment as a principled mitigation. Label efficiency is also rarely addressed jointly with robustness under asynchrony, despite beam labels being fundamentally expensive in the target deployment context.

**Our contributions in context.** This work fills the identified gap by unifying three elements in a single end-to-end pipeline: (i) delay-regime-aware asynchronous alignment that treats staleness and missingness as first-class variables, supported by the staleness-induced beam misalignment bound of Section III-A; (ii) reliability-guided fusion with uncertainty-grounded alignment regularization that prevents noise propagation at the alignment stage, supported by the gradient contamination bound of Section III-B; and (iii) label-efficient two-stage learning with degradation-aware MoE-LoRA adaptation, with a complexity analysis confirming real-time feasibility (Section IV). The proposed evaluation suite—including tail-risk metrics and systematic ablations A1–A5—makes every contribution claim independently falsifiable under deployment-realistic conditions.

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